

Number theory – 1

Congruences: Fermat, Euler and the Chinese remainder theorem

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Theoretical minimum. Everything is proved here; no course in number theory is presupposed. The group-theoretic reading of the same facts is Abstract algebra – 1.

Definitions and the Euclidean algorithm. For integers a, b not both zero, $\gcd(a, b)$ is the least positive element of $\{ax + by\}$; the Euclidean algorithm computes it and produces x, y with $ax + by = \gcd(a, b)$ (Bézout). Consequently $p \mid ab$ implies $p \mid a$ or $p \mid b$, and factorisation into primes is unique. We write $a \equiv b \pmod{n}$ when $n \mid a - b$; the residues form the ring $\mathbb{Z}/n\mathbb{Z}$, whose invertible elements are the residues coprime to n , $\varphi(n)$ in number.

Theorem 1 (Chinese remainder theorem). If $n = n_1 n_2 \cdots n_k$ with the n_i pairwise coprime, then the natural map

$$\mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

is an isomorphism of rings. *Proof:* it is a homomorphism, both sides have n elements, and it is injective because $n_i \mid x$ for all i forces $n \mid x$. *Consequences:* φ is multiplicative, so $\varphi(n) = n \prod_{p \mid n} (1 - 1/p)$; a system of congruences with pairwise coprime moduli is always solvable; and any question about $\mathbb{Z}/n\mathbb{Z}$ splits into questions about prime powers. This splitting is the single most used device of the whole section.

Theorem 2 (Fermat, Euler, Wilson). If $\gcd(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod{n}$, and for prime p , $a^p \equiv a \pmod{p}$. *Proof:* multiplication by a permutes the invertible residues, so the product of all of them is unchanged; alternatively this is Lagrange's theorem for the group $(\mathbb{Z}/n\mathbb{Z})^*$. The same

product argument gives Wilson's theorem $(p-1)! \equiv -1 \pmod{p}$, the pairing of each residue with its inverse leaving only ± 1 unpaired.

Theorem 3 (lifting a solution: Hensel). Let $f \in \mathbb{Z}[x]$ and suppose $f(a) \equiv 0 \pmod{p^k}$ with $f'(a) \not\equiv 0 \pmod{p}$. Then there is a unique $b \equiv a \pmod{p^k}$ with $f(b) \equiv 0 \pmod{p^{k+1}}$. *Proof:* write $b = a + tp^k$ and expand, $f(a + tp^k) \equiv f(a) + tp^k f'(a)$, which is a linear congruence in t with invertible coefficient. Thus a simple root modulo p lifts to every power of p , and the only genuine obstructions live modulo p and at the multiple roots.

Fact 1 (the modular obstruction). To prove that an equation has no solutions in integers, exhibit a modulus for which it has none: squares are $0, 1 \pmod{4}$ and $0, 1, 4 \pmod{8}$; cubes are $0, \pm 1 \pmod{9}$; fourth powers are $0, 1 \pmod{16}$. The converse fails — solvability modulo every n does not imply solvability in $\mathbb{Z}$ — and the classical counterexample is $(x^2 - 2)(x^2 - 17)(x^2 - 34) = 0$.

1. Prove Bézout's identity, the uniqueness of prime factorisation, the Chinese remainder theorem in the ring form and the multiplicativity of φ . Then prove Euler's and Wilson's theorems, and deduce that p is prime if and only if $(p-1)! \equiv -1 \pmod{p}$.
2. Find all positive integers $n < 10^{100}$ for which simultaneously $n \mid 2^n$, $n-1 \mid 2^n - 1$ and $n-2 \mid 2^n - 2$. (Putnam 2018)
3. Let n be a positive integer. Find the number of solutions of $x^2 \equiv x \pmod{n}$ in $\mathbb{Z}/n\mathbb{Z}$, and explain the answer through the Chinese remainder theorem and idempotents. (ISTCiM 2022)
4. Let f be a polynomial of degree 2 with integer coefficients such that $f(k)$ is divisible by 5 for every integer k . Prove that all coefficients of f are divisible by 5, and determine for which degrees and primes the analogous statement holds. (IMC)
5. Let $A = \{n : 1 \leq n \leq 2021, \gcd(n, 2021) = 1\}$ and $S(j) = \sum_{n \in A} n^j$

for $j \geq 0$. Determine all j for which $S(j)$ is a multiple of 2021. (Putnam 2021)